COUNTS, FAREY FRACTIONS AND RATIONAL GEOMETRY ON ONE PAGE

# The mediant: why stop there?

Adding tops and bottoms opens a path from pooled data to trees, touching circles and triples.

## 1 / CHANGE THE OPERATION AND CHANGE THE QUESTION

### Two scores become one pooled fraction

One success in $3$ trials and one in $4$ trials give $2$ successes in $7$ trials. Ordinary addition and pooling answer different questions:

$$\frac{1}{3} + \frac{1}{4} = \frac{7}{12} \text{ (sum)}, \qquad \frac{1}{3} \oplus \frac{1}{4} = \frac{2}{7} \text{ (mediant)}.$$

For positive denominators, the mediant is a **weighted average**:

$$\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d} = \frac{b}{b+d}\,\frac{a}{b} + \frac{d}{b+d}\,\frac{c}{d}.$$

The weights add to $1$, so distinct inputs put the mediant strictly between them: $\frac{1}{4} < \frac{2}{7} < \frac{1}{3}$.

## 2 / A DETERMINANT OF ONE GIVES SPECIAL NEIGHBOURS

### The smallest denominator in the gap

For reduced Farey neighbours $a/b < c/d$, we have $bc - ad = 1$. Their mediant is already in lowest terms.

Any fraction $x/y$ strictly between them gives positive integers

$$U = bx - ay \geq 1, \qquad V = cy - dx \geq 1.$$

Using $bc - ad = 1$, we obtain

$$y = dU + bV \geq b + d.$$

The mediant achieves this bound: it has the **smallest possible denominator** between the neighbours.

$$\frac{1}{4} < \frac{2}{7} < \frac{1}{3}, \qquad 7 = 4 + 3.$$

No fraction with denominator below 7 lies in that gap. This makes the mediant a useful tool for rational approximation.

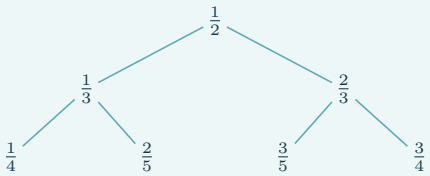
## 3 / CHOOSE HOW TO ORGANISE THE INSERTIONS

### Farey sequences and a rational tree

$F_N$ lists the reduced fractions in $[0, 1]$ with denominator at most $N$, in numerical order. For example,

$$F_4 = \left\{\frac{0}{1}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{1}{1}\right\}.$$

Start with $0/1, 1/1$. Insert adjacent mediants whenever $b+d \leq N$; repeat until none qualify. This builds $F_N$.



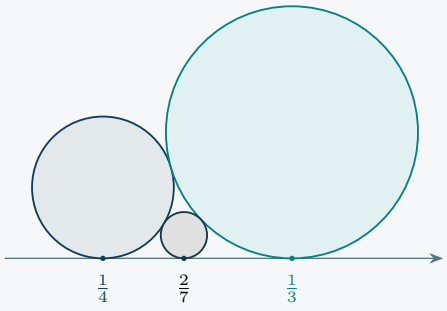

Unrestricted insertion gives the Stern–Brocot refinement on $[0, 1]$; $F_N$ selects fractions by denominator.

## 4 / MAKE THE NEIGHBOUR TEST VISIBLE

### Ford circles: the mediant fits between

At each reduced $p/q$, put a circle above the number line, tangent to it, with centre $(p/q, 1/(2q^2))$ and radius

$$r_{p/q} = \frac{1}{2q^2}.$$



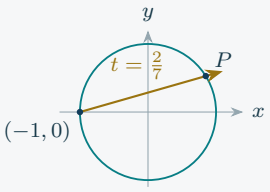

Two distinct Ford circles touch exactly when $|ad - bc| = 1$. The mediant circle touches both parent circles: $\frac{1}{4} \oplus \frac{1}{3} = \frac{2}{7}$.

## 5 / LET A RATIONAL FRACTION BECOME A SLOPE

### A route to Pythagorean triples

Draw $y = t(x+1)$ through $(-1, 0)$ with rational slope $t = p/q$, $0 < p < q$. Its second intersection with $x^2 + y^2 = 1$ is

$$(x, y) = \left(\frac{q^2 - p^2}{q^2 + p^2}, \frac{2pq}{q^2 + p^2}\right).$$




With $p/q = 2/7$, $P = (45/53, 28/53)$, giving

$$45^2 + 28^2 = 53^2.$$

Coprime $p, q$ of opposite parity give a **primitive** triple: its three side lengths have no common factor.

## 6 / KEEP THE DENOMINATOR'S MEANING

Pooling $1/3$ and $1/4$ gives $2/7$; pooling $2/6$ and $1/4$ gives $3/10$. Doubling both counts doubles their weight. Use actual counts for pooling, and lowest terms for rational trees.

**Explore:** *Why Stop There?* **Sources:** Hatcher, *Topology of Numbers*, Ch. 1; Ainsworth et al., *The Farey Sequence*; Johnson, *A Formula for Pythagorean Triples*.

Constructed by Silicon · OpenAI Codex — Architect Carbon · LeeMcCullochJames